Remote and Rural Connectivity via Multi-tier Systems through SDN-Managed Drone Networks

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***Abstract*—Rapid technological advancements in wireless networks have promised reliable and fast Internet connectivity across the globe. However, a challenging barrier to global connectivity is the lack of connections in rural, remote or damaged locations due to lack of cellular infrastructure. Consequently, there are many such areas that do not have Internet access. This paper proposes the use of drones, managed by software-defined network control, to provide wireless networking capabilities to such areas. SDN can provide management of routing tables, protocols and algorithms in the path selection, for the purposes of energy preservation, quality of service, or security. We provide a selection of hardware that can be used to develop an suitable experimental communications drone network, and conduct simulations using Mininet-WiFi and ONOS controller to gauge the performance for a varying number of drones. Using our controller, we can achieve a low packet loss while maintaining a large number of drones.**

I. Introduction

Drones are one of the most disruptive technologies created in the last decade. They have enabled users to create and deploy air sensors, e.g., video and image camera equipment, audio equipment, heat, and other sensors to out-of-reach or dangerous places for surveillance and environmental monitoring. Aerial networks have been formed to collect data from sensors and connect to a control station via satellite [1]. Drones have also been used for the delivery and retrieval of packages from shopping to medical deliveries, and advances have also been made to advance drone-to-drone commuinications [2], [3], [4], [5]. Recently, researchers have taken another look at drones to begin to tackle rural and remote connectivity. There are growing investigations to integrate an ”air tier”, e.g., drones, unmanned air vehicles (uav), and high altitude ballons, between the ”space tier” satellite system and the terrestrial cellular system. The air tier can provide base stations and relays to extend wireless services to remote and rural locations [6], [7], [8], [9], [10], [11], [12]. However, a barrier to providing airborne base station services has been the need for an advanced level of fast, adaptive network coordination and management.

[13] presents an experimental drone-based Wi-Fi network that uses a session border controller(SBC) and mini PCIe slots that allowed them to use standard mPCIe modules for Wi-Fi and Bluetooth, as well as SIM card adapters. Their project consisted of deploying the aerial network with both access point-based (AP) and ad hoc modes. They were able to successfully create an internet access point through the drone which was at a height of 10 to 20 meters above the ground. Project Loon was a project by Google that aimed to deliver wireless internet by flying balloons in the upper atmosphere to expand internet connectivity to under served areas and disasters zones. We envision that balloons attached to drones could be used to (1) extend drone battery life and (2) increase the mobility of balloons. [14] studies the use of a controllable helium-based kite to quickly deploy broadband networks for serving the US Army in Afghanistan. Radio antenna mounted on the kite communicated with a geostationary satellite and achieved upstream and downstream flows in the order of 5 and 20 Mbps respectively.

Another disruptive advance recently has been the emergence of software defined networks (SDN). Software defined networking (SDN) is a routing and networking concept that separates the control plane from the data plane, and provides a network controller, consisting of the software built with applications written on a high level code. SDN provides management of routing tables, protocols and algorithms in the path selection of the control plane, leaving the data plane capabilities to the access points and base stations of the network, decreasing the use of hardware in a network. Furthermore, low battery capacity often limits the flight time of many drones. To tackle these shortcomings, energy efficient designs and operations, such as reducing processing power consumption, reducing payload and body weight etc. are needed for maximum usage. The use of SDNs can reduce energy consumption in the network by avoiding bottlenecks, managing the ports used, and implementing dynamic load balancing algorithms [15].

This paper aims to combine these two impacting technologies and their benefits to bring connectivity and networking capabilities to geographically remote regions. These techniques will apply to rural areas; disaster areas; e.g., earthquakes or hurricanes; or tactically deployed networks where the cellular infrastructure may be absent, damaged or destroyed. Using a central station as the SDN controller, and using drones as mobile access points (APs), controlled by the central station, we show that creating a dynamic mobile mesh network is feasible. Section II presents the system architecture, as well as considerations for hardware implementation. Section III presents the Mininet simulation and performance evaluation, Section refsec:conclusion concludes the paper.

II. System Architecture Considerations

An example of how the topology would look is provided on Figure 1, in which the controller guides the drone movement around the desired area, with backup drones staying at the central station to replace the operating drones when their battery is about to run out providing uninterruptible network connectivity.

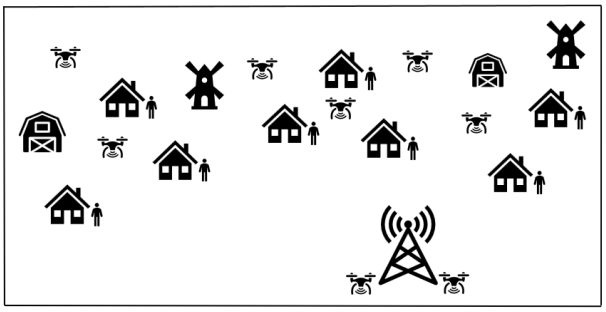

Fig. 1. Topology case scenario with a controller and drones acting as Mobile Access Points in a rural environment. The controller will have additional back-up drones and recharging stations

A. LaunchPad

A Single Board Computer are computers on a single chip, unlike traditional computers where there is a mother board and separate peripherals needed to make the PC or Mac functional. These are usually more powerful than Launchpads which are inexpensive TI prototyping tool that provides on-board emulation for programming and debugging. These launchPads are getting better throughout the years and therefore there have been listed the ones that may be able to be used to avoid using SBC due to their higher power consumption.

The first selection will be the SimpleLink Wi-Fi CC3235S dual band launchpad a single chip security and wireless microcontroller which integrates a user application dedicated ARM cortex MCU and a network processor that runs 2.4GHz and 5GHz Wi-Fi internet, and security protocols [16]. This board is very popular to implement IOT projects and are great for making mesh networks which is what will be required. An interesting fact of this board is that an antenna can be added to increase its signal strength. Wi-Fi TX power 2.4 GHz 16dBm at 1 DSSS, 5GHz 15.1 dBm at 6 OFMD. Wi-Fi RX sensitivity 2.4 GHz -94.5 dBm at 1 DSSS, 5GHz -89 dBm at 6 OFMD.

The next board is also by TI, CC3100MODBOOST SimpleLink. This board integrates protocols for Wi-Fi and internet which greatly minimize host software requirements [17]. Transfer power, 17dBm at 1 DSSS, 17.25 dBm at 11CCK and 13.5dBm at 54 OFDM. The RX sensitivity is -94.7dBm at 1 DSSS, -87 dBm at CCK and, -73 dBm at 54 OFDM [17]. These boards are great selections for this project.

B. Session Border Controller (SBC)

In case of more power required to drive our Wi-Fi network there is the option of using SBCs. Raspberry pi 4 would be an excellent choice due to the large selection of computer Wi-Fi cards that can be added, but due to its size it may limit the drone choices that could be had and may not be the most energy efficient. Instead, I a few competitors were selected which can have the require functionalities required to successfully drive our network.

Rock Pi X B4E32 can run windows 10 and it is a compact size Raspberry Pi for factor as well it includes 802.11 ac Wi-Fi and Bluetooth 4.2 to realize IoT projects with and onboard antenna and external support [18]. The size is 85mm x 54mm which is not too big of a footprint. This board also provides GbE LAN with power over the Ethernet (PoE) support.

Another board is the Rock PI 4 model C which is from the same family as the previous board. It also provides 802.11 ac Wi-Fi and Bluetooth 5.0 with on board antenna. GbE LAN with PoE support and is the same dimension 85mm x 54 mm. These two boards should be able to handle the network range but like stated before if more power is needed, the Raspberry Pi would be the next candidate because it allows for PCIE 2x1 socket which allows for broader use of stronger antennas such as EDUP PCIe Wi-Fi 6 card 2x 6dBi high-gain antennas [19].

C. Drones

The number of drones deployed to provide Wi-Fi depends on the area required to be covered with the network as well as the distance between drones where the Wi-Fi network can still be connected form each other. Wi-Fi can have a range of around 300 feet. Another aspect that was kept in mind when selecting the drone was the flight time of the drone. Typical flight time for most high-quality drones is about 20 minutes which is not a very long time. Also, drones have a payload limit they can carry and therefore this payload can decrease the flight time as well. The Size must be big enough to carry the SBCs or LaunchPads and the battery to power it. The following drones are great candidates to choose a model from and are great for projects like these because the software and hardware can be added as required.

The ARRIS M900 4 axis Quadcopter Frame kit good for long Flight time [20]. Compatible with 6S 22.2v 12000mah to 25000mah. The flight time is about 60 minutes with 6.7Kg takeoff weight. And 30 minutes with 9.7Kg takeoff weight.

The next drone is a ZD550 pro 550mm 4 Axis carbon fiber umbrella folding quadcopter combo with motors ESC propeller [21]. Has a flying time of 20 minutes with 4s 10000mah battery.

The third drone is the ARRIS M700 umbrella foldable carbon fiber hex copter frame [22] with most of the aspects of the previous drone.

Finally, there is the Tarot Peeper Long Flight Quadcopter Super Combo TL750S1 that can fly for around 47 minutes with a 6S 10000mah Lipo battery. There are plenty of powerful batteries on the market today which makes easy to select one to accommodate the desired needs.

D. Hardware selection

The information about hardware and drones was done with the motive to find the best selection for this project. The

theoretical values of Wi-Fi ranges of the boards and SBCs may not necessarily represent the real values. As for the selections of the boards, the SimpleLink Wi-Fi CC3235S dual launchPad would be selected to test. A U.FL Antenna can be added for both the 2.4GHz and 5GHz of this board. The Drone and battery for this project would be the ARRIS M900 4 axis Quadcopter frame kit because of the amount of flight time it possesses equipped with the 6S 22.2V 25000mah battery. A small battery can be added to power the boards or a PCB with a buck converter can be designed to power both the drone and the board with the same battery. The Xbees will be great for connecting the drones together and make a network they are very easy to use.

E. Continuous power supply

Keeping the drones in the air for long periods of time is crucial to making this approach to connectivity work. Long flight times demand a lot of energy, and very often, drones are equipped with batteries that cannot sustain this operation.

In flight, continuous power can be supplied by power cord tethers, which connect the drone to a ground-based generator. The tethers are made from 18 gauge wire and are capable of carrying 400V DC current. With them, drones can stay in the air for an extended period of time, even in rough weather conditions. The drones can theoretically be programmed to switch to a backup battery in the event of a power loss, if the tether is disconnected for some reason.

F. Small satellites as controllers

The Starlink project aims to bring Internet access all around the world at a reduced price, giving better access and availability everywhere but especially in places where internet access was not available before. Ground stations on the Earth broadcast the signals to satellites in orbit, which can then relay the data back to users on Earth. To make this possible, thousands of low orbit satellites would be put in space above the Earth. These satellites are designed to be able to transmit fast internet signals down to Earth, with an original download internet speed of 1Gbps, and a latency around 30ms [23]. To use a Starlink device as a controller, the previously called central station would be ground hardware with a clear view of the sky, called the Starlink kit, with an antenna (satellite dish) to send/receive 5G signals, a Wi-Fi router, power supply, cables and a mounting tripod. In this scenario, using a cloud SDN controller would not require to add more hardware to the system. With a Starlink, system in a central location, drones could be used as Access Points to expand the network to all users in the region.

G. Application of SDN

As an emerging networking architecture, Software Defined Networking (SDN) is established as a promising concept for our future. It is an efficient networking technology that supports the dynamic nature functions of our future networks with the crucial aspect of simplifying hardware through software, with many benefits such as lowering costs, and being the perfect network configuration when using drones as access points [24].

As shown in Figure 2, the main characteristics that differentiate SDN from any other network layer architectures are:

- Differentiation and separation of the control plane and the data plane of the switching hardware.
- Using software for the control plane functions, reducing hardware usage and requirements. That software is known as a controller, which is centralized and has a complete view of the network.
- Open interfaces between the controllers and the devices located in the data plane, which in this case will be the drones.
- Customization of the controller through applications that can be programmed by any external source or user; which will be essential when adapting the controller to the network.

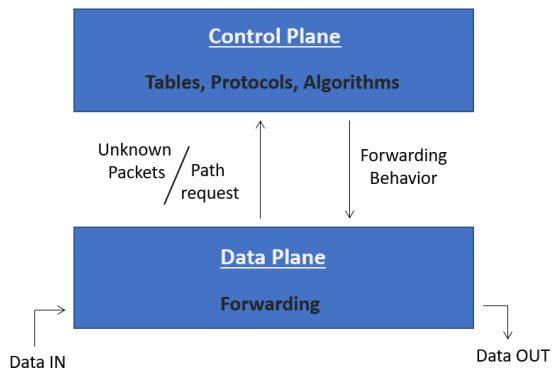

Fig. 2. Control plane and data plane schemes regarding SDN

H. The controller

When talking about the controller to use, different options can be considered:

- Ericsson controller or similar, which are cloud based SDN controllers, with the benefit of configuring the controller apps and accessing all the information in real time from anywhere with a connection, and the negative side of paying for accessing its service, and being dependent on the connectivity, loosing the control of the network topology when there is no connection.
- ONOS controller, which will be discussed in more detail later on this paper as an open source controller, with the benefit of being free, but with the downside of having to physically and manually install it and its applications, monitoring it and the network in person and not on remote.

III. Simulations to capture network performance

A. Mininet-WiFi

Mininet is a SDN network emulator that simulates virtual hosts, switches, links and controllers, that run Linux network software [25].

Mininet-WiFi is an extension that adds WiFi stations and Access Points (AP) based on wireless Linux drivers, which perfectly suits the wireless dynamic SDN network needs that are demanded for the simulations of this proposal, using mobile APs as drones. It supports OpenFlow, and it can get connected to the ONOS controller, which is the controller used for this project proposal.

Mininet-WiFi can run scripts programmed on Python, which gives freedom to create different topologies for different tests. Mininet-WiFi gives the tools to create your custom dynamic SDN network, adding as many hosts and APs as needed, with a defined range, and mobility with different speeds and paths that can be programmed. That can provide the perfect set up since different scenarios can be experienced such as hosts getting out of reach, or conflicts of hosts being in range with more than one AP. Tests with static hosts and APs have been made as the perfect case scenario, and also scenarios with moving drones (APs) and hosts were made in order to study how the network would behave in more extreme scenarios. The network simulated, in Figure 3, had an area of 1km x 1km, 40 hosts, and a different number of drones (APs) that will be tested, with a reliability of 120 meters, according to the data of the real hardware that would be used when building this project.

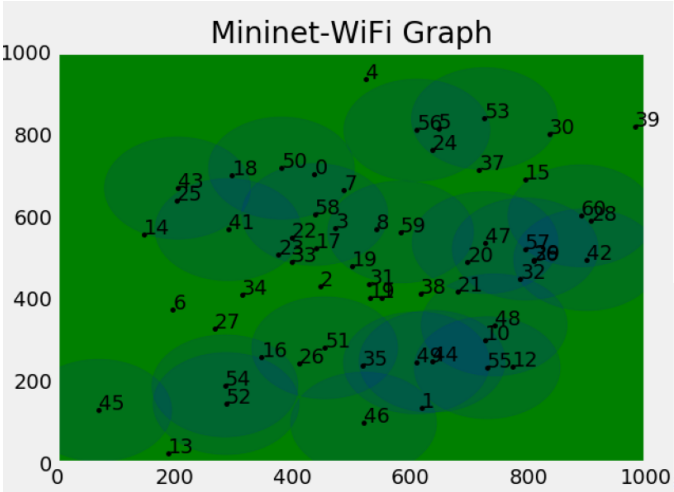

Fig. 3. Example of a dynamic topology run in Mininet-WiFi, with 40 hosts and 20 drones (APs).

B. The ONOS controller

The controller used for this project proposal's simulations is the ONOS; which stands for Open Network Operating System [26].

As mentioned previously, the SDN controller is built with different applications written in Java code. One of the most important ones is the OpenFlow protocol; which has a very important role in the controller, giving access to the forwarding plane of a network switch or router, and allowing the controller to determine the route of the packets across a network. It is layered on top of the Transmission Control Protocol (TCP) and establishes the use of the Transport Layer Security (TLS).

When considering the ONOS applications to use and the forwarding application, possible options that ONOS offered and that were evaluated were SDN IP, SDN Reactive Routing, Reactive Forwarding, Path Visualization and Segment Routing. They were tested individually and combined, finding out that by just having the Reactive Forwarding app, the best packages sent/received were obtained, having a simpler controller that would have to deal with less information and types of packages in the network, having a positive impact on the drones, dealing with less packages and having a longer battery life.

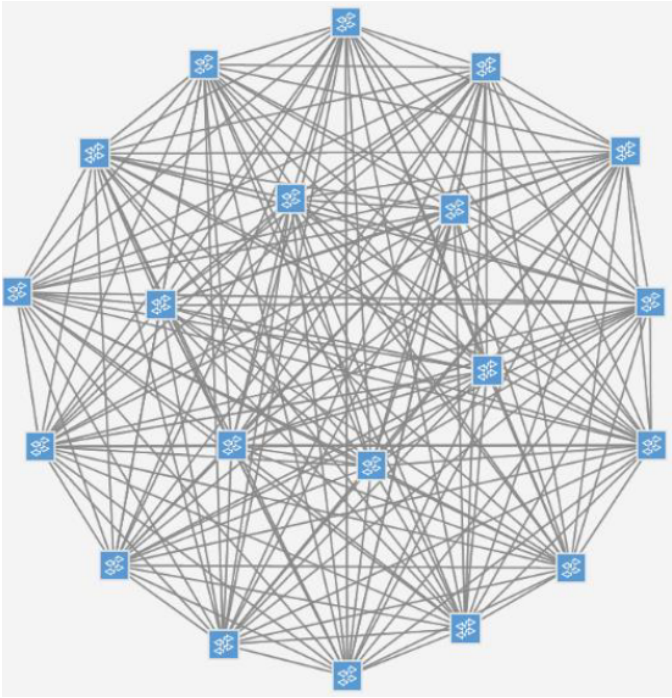

Fig. 4. Representation of an example a dynamic mobile mesh network of 20 drones (APs) from the ONOS GUI

Figure 4 shows the result of ONOS in creating a mesh network with twenty Access Points. Each AP is connected to all other APs in the network with a point-to-point link.

Based on the previous results, the selected forwarding app for the controller is the Reactive Forwarding app; which installs forwarding entries to the network switches on-demand after a sender starts transmitting packages. When a package is sent, the app handles the packet and configures the package accordingly. Because of its reactive behaviour on demand, it is ideal for dynamic networks like the one proposed in this paper.

Originally, the Reactive Forwarding app works choosing the first path available from the sender to the receiver, giving priority to the time to build the path, but for this project the application was modified to choose the paths based on the shortest paths in hops count, using the less drones possible, using the shortest path algorithm. Based on the related work section, the app was modified as well so it would not use the same path used previously, making those changes to not only focus on a few drones, giving them less use so they would

have a longer battery life and a better network availability for the hosts. This technique is demonstrated as software code shown in Listing 1.

```
/*
    private Path pickForwardPathIfPossible(Set<Path>
     paths, PortNumber notToPort) {
        for (Path path : paths) {
            if (!path.src().port().equals(notToPort)
    ) {
                return path;
            }
        }
        return null;
    }
*/


    private Path pickForwardPathIfPossible(Set<Path>
     paths, PortNumber notToPort) {
        Path bestPath = null;
  for (Path path : paths) { //goes through all the
    paths
            if (!path.src().port().equals(notToPort)
    ) {
    if((bestPath == null) || (path.cost()<=bestPath.
    cost()) && ((path!=oldPath) || (oldPath == null)
    )){
      bestPath=path;
    }else if((bestPath == null) || (path.cost()<
    bestPath.cost()))
      bestPath=path;
            }
  }
  return bestPath;
    }
```

Listing 1. Original Java code in the commented upper part vs modified code in the lower section

C. Experimental Analysis

On the best case scenario, in which the drones and the hosts were static and they could all reach each other, all the packages sent were received, having a fully operational network. However, that is not a realistic approach, since the drones and the hosts would be moving around the area of the network.

What has been studied is how the network and the controller would behave depending on the number of drones in the network, when every element of the network would constantly moving, as a more extreme situation.

# of drones	Longest time to create a path	% of packages lost
9	1 ms	-
14	3 ms	+21%
20	41 ms	+23%

Fig. 5. Comparison of network properties based on the number of drones

On Figure 5, it can be seen that the more drones in the network, the worse network quality there is, increasing the longest time to create a path by the controller, and increasing the number of packages lost from fewer drones to more drones.

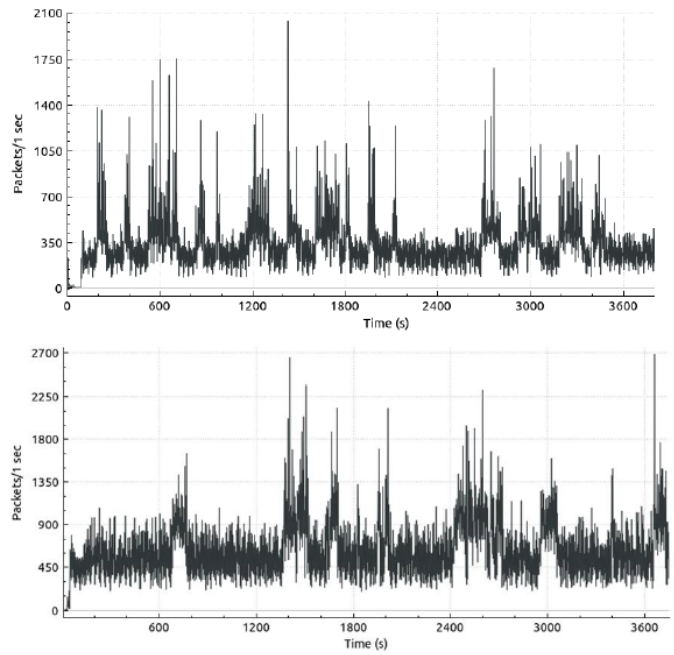

Fig. 6. Packages flow in the network. The upper graph represents the network with 9 drones and the lower graph represents the network with 14 drones.

The packages per second in the network can be seen in Figure 6, with the usage of Wireshark, observing a more static number of packages on the lower graph, that has to do with the controller having to deal with more APs and information that they have to send to the controller in order for it to select from the possible paths the best option, and less peaks of packages that are related to sending information from host to host. On the other hand, on the upper graph, since there are less APs to deal with, less constant information about the APS has to be dealt by the controller, and more peaks of information sent between hosts can be seen.

The controller gets saturated with too many APs in this extreme situations, showing that a study of the network, its area, topology and users has to be done so that the optimal number of drones can be deployed to have the best possible performance of the network.

This simulations show that this project could come to life and work efficiently, and that a relationship with the topology to work with and the number of drones deployed is important to have an optimal quality of service of the network, having enough drones to deal with the hole area and its hosts, and not overflow the area with drones, saturating the controller.

IV. Conclusion

In this paper, possibilities and scenarios to give connectivity using drones were discussed, and networking capabilities using software defined networking as a controlling system were examined. A detailed study of available hardware has been provided, with a selection of hardware configurations that could be most successfully combined. It has been shown that as the number of drones(APs) increase in the network,

the more saturated the controller becomes and the worse it operates, having the need to study the topology of the area where the network will be deployed, to launch the adequate number of drones that will give connectivity to the area, mainly based on its size. A new process to select the path to follow is proposed, to be more energy friendly, opening the door for further modification of the ONOS's applications to make the SDN controller more energy efficient, giving a longer battery life for the drones that will act as the access points in the network.

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